Clinical Risk Assessment Report: Patient 87

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: High Confidence
- Absolute Predicted Risk: 45.0% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 2.03x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM_LGZE.CT), which elevated the patient's absolute risk by +7.5%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.33 [Bottom 19%] (+7.5%)**

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 0.2 [Avg (60th %ile)] (+4.4%)**

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.68 [Avg (67th %ile)] (+2.7%)**

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: -0.06 [Top 15%] (+2.2%)**

Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

Short Run High Gray-Level Emphasis (CT) (GLRLM_SRHGE.CT) **Value: 0.73 [Avg (72th %ile)] (+2.2%)**

Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.22 [Bottom 14%] (+2.0%)**

Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Patient Age (Age) **Value: 57.0 [Avg (27th %ile)] (+2.0%)**

Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Tumor Local Contrast (PET) (GLCM_Contrast(=Variance).PET) **Value: -0.4 [Avg (47th %ile)] (+1.8%)**

Measures local variations and sharp transitions in density or metabolism, signifying an unstructured and variable tumor matrix.

3. Protective / Mitigating Factors (Reducing Risk)

No major risk-reducing features observed in the top drivers.

Machine Learning Feature Attribution (SHAP Decision Path)